

Medicinal Actions & Uses A slightly bitter-tasting remedy, lady's bedstraw is used mainly as a diuretic and for skin problems. Like its close relative, cleavers (*G. aparine*, preceding entry), the herb is given for kidney stones, bladder stones, and other urinary conditions, including cystitis. It is occasionally used as a means to relieve chronic skin problems such as psoriasis, but, in general, cleavers is preferred as a treatment for this condition. Lady's bedstraw has had a longstanding reputation, especially in France, of being a valuable remedy for epilepsy, though it is rarely used for this purpose today.

Related Species *G. elatum* has also been considered a remedy for epilepsy in France. (See also *G. aparine*, preceding entry.)

Gardenia jasminoides syn. *G. augusta*, *G. florida* (Rubiaceae) Gardenia, Zhi Zi (Chinese)

Description Evergreen shrub growing to 6½ ft (2 m). Has green leaves, scented double flowers, and orange-red fruit.



Gardenia plays a significant role in Chinese herbalism.

Habitat & Cultivation Native to southeastern provinces of China, gardenia prefers humid, tropical climates. The fruit is gathered when it turns reddish-yellow.

Part Used Fruit.

Constituents Gardenia fruit contains iridoid glycosides. The flowers are the source of the essential oil.

History & Folklore Gardenia has been used in Chinese medicine for at least 2,000 years. It provides an important essential oil used to flavor teas. The oil is also used to make perfumes. Gardenia perfumes often blend gardenia, jasmine, and tuberose.

Medicinal Actions & Uses In the Chinese herbal tradition (pp. 40–43), gardenia is a “bitter,

cold” herb used mostly to relieve symptoms associated with heat. These include fever, irritability, restlessness, insomnia, cystitis, painful urination, and jaundice. The fruit staunches bleeding and is taken for nosebleeds, and urinary and rectal bleeding. The essential oil is an antiseptic and has a reputation for being an aphrodisiac.

Related Species The fruit of the northern Indian *G. campanulata* is cathartic and used to expel worms. *G. gummifera*, from eastern India, is antiseptic and digestive. The Pacific region *G. taitensis* relieves headaches. The African *G. thunbergia* is used to relieve constipation.

Cautions Avoid in diarrhea. Do not take if taking prescribed medication for high blood pressure.

Gaultheria procumbens (Ericaceae) Wintergreen

Description Aromatic low-lying shrub growing to 6 in (15 cm). Has leathery, oval leaves, small white or pale pink bell-shaped flowers, and brilliant red fruit.

Habitat & Cultivation Native to North America, wintergreen is found in woodland and exposed mountainous areas. The leaves and fruit are gathered in summer.

Parts Used Leaves, fruit, essential oil.

Constituents Wintergreen contains phenols (including gaultherin and salicylic acid), 0.8% volatile oil (up to 98% methyl salicylate), mucilage, resin, and tannins.

History & Folklore Wintergreen was popular with Native Americans, who used it for treating back pain, rheumatism, fever, headaches, sore throats, and many other conditions. Samuel Thomson, founder of the 19th-century Physiomedicalist movement, combined it with hemlock (*Conium maculatum*, p. 193) to treat severe fluid retention. The leaves have been used as a substitute for tea (*Camellia sinensis*, p. 181), for example during the American Revolutionary War (1776–1784).

Medicinal Actions & Uses Wintergreen is strongly anti-inflammatory, antiseptic, and soothing to the digestive system. It is an effective remedy for rheumatic and arthritic problems, and, taken as a tea, it relieves flatulence and colic. The essential oil, in the form of a liniment or ointment, brings relief to inflamed, swollen, or sore muscles, ligaments, and joints, and can also prove valuable in treating neurological conditions such as sciatica (pain resulting from pressure on a nerve in the lower spine) and trigeminal neuralgia (pain affecting a facial nerve). The oil is sometimes used to treat cellulitis, a bacterial infection causing the skin to become inflamed.

Cautions People who are sensitive to aspirin should not take wintergreen internally. Oil of wintergreen should never be taken internally,

nor applied (even well diluted) to the skin of children under the age of 12 unless with professional supervision.



Wintergreen makes an effective liniment for sore muscles and joints.

Gelidium amansii (Rhodophyceae) Agar

Description Seaweed with red-brown, translucent, multi-branched ribbons and fronds growing to about 3 ft (1 m) in length. It has spherical fruit that appears in late autumn and winter.

Habitat & Cultivation Agar is native to the Pacific coasts of China and Japan and the coast of South Africa. It grows to a depth of 100 ft (30 m) below sea level. Commercial harvesters rake plants off banks and rocks. The cleaned seaweed, after being boiled with sulphuric acid for 6 hours, yields agar, which sets to form a jelly. Around 6,500 tons of processed agar are produced each year.

Part Used Seaweed extract (agar).

Constituents Agar contains polysaccharides, mainly agarose and agaropectin (up to 90%), which are very mucilaginous.

History & Folklore Agar is commonly used as a thickening agent in food preparation, but its most widespread application is in scientific research, where it is used as a culture medium for growing micro-organisms in petri dishes.

Medicinal Actions & Uses Like most seaweeds and their derivatives, agar is nutritious and contains large amounts of mucilage. Its chief medicinal use is as a bulk laxative. In the intestines, agar absorbs water and swells, stimulating bowel activity and the subsequent elimination of feces.

Related Species While *G. amansii* is the main agar-producing species, *G. cartilagineum* (found on the Pacific coast of North America) and other closely related species around the world are being used as alternative sources.